NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Electronics and Communication Engineering
MID SEMESTER EXAMINATION, July-Dec 2026
Branch: B. Tech EE 3rd Sem VLSI(Electronics)

Course Name: Analog Electronics **Course Code: EC031302**
Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. ***Attempt all questions.***

	Questions Description	Marks
Q1.	**(a)** Sketch the frequency response of single-stage and multistage RC-coupled amplifiers. Explain the gain roll-off at low and high frequencies and show why the phase shifts of a CE amplifier are 225° at f_L and 135° at f_H, respectively. **(b)** Match Column A with the correct expressions in Column B as shown Table 1.	3+3+4

Table 1: Amplifiers

Column A – Amplifier	Column B – Parameters
A. CE with Bypass Capacitor	1. $A_I = \frac{h_{fe}}{1+h_{fe}}$; $R_i = \frac{h_{ie}}{1+h_{fe}}$; $A_V = \frac{h_{fe}R'_L}{h_{ie}}$; $R_o = \infty$
B. Common Collector (CC)	2. $A_I = -h_{fe}$; $R_i = h_{ie} + (1+h_{fe})R_E$; $A_V = -\frac{h_{fe}R'_L}{h_{ie}+(1+h_{fe})R_E}$; $R_o = \infty$
C. Common Base (CB)	3. $A_I = -h_{fe}$; $R_i = h_{ie}$; $A_V = -\frac{h_{fe}R'_L}{h_{ie}}$; $R_o = \frac{1}{h_{oe}}$
D. CE without Bypass Capacitor	4. $A_I = \frac{1+h_{fe}}{1+h_{oe}R_E}$; $R_i = h_{ie} + (1+h_{fe})R_E$; $A_V = \frac{(1+h_{fe})R_E}{h_{ie}+(1+h_{fe})R_E}$; $R_o = \frac{h_{ie}+R'_s}{1+h_{fe}}$
E. Darlington Amplifier	5. $A_I = -h_{fe}$; $R_i = h_{ie}$; $A_V = -\left(\frac{h_{fe}}{h_{ie}}\right)R'_L$; $R_o = \infty$; $g_m = \alpha_2 g_{m1}$
F. Cascode Amplifier	6. $A_I = (1+h_{fe})^2$; $A_V \approx 1$; $R_o \approx \frac{h_{ie}}{1+h_{fe}}$ (tens of Ω) ; R_i not specified

Column A – Amplifier	Column B – Parameters
A. CS, Resistive Load	1. $A_V = -g_{m1}(r_{o1} \parallel r_{o2})$; $R_{in} = \infty$; $R_{out} = r_{o1} \parallel r_{o2}$
B. CS, Diode Load	2. $A_V = \frac{(1+g_m r_o)R_D}{r_o + R_D}$; $R_{in} = \left(\frac{1}{g_m}\right) \parallel r_o$; $R_{out} = r_o \parallel R_D$
C. CS, Triode Load	3. $A_V = -g_m(r_d \parallel R_D)$; $R_{in} = \infty$; $R_{out} = r_d \parallel R_D$
D. CS, Ideal Current Source	4. $A_V = \frac{g_m(r_o \parallel R_S)}{1+g_m(r_o \parallel R_S)}$; $R_{in} = \infty$; $R_{out} = \left(\frac{1}{g_m}\right) \parallel r_o \parallel R_S$
E. CS, PMOS Current Source	5. $A_V = -g_{m1}r_{o1}$; $R_{in} = \infty$; $R_{out} = r_{o1}$
F. Common Drain	6. $A_V = -g_{m1}(r_{o1} \parallel R_{out2})$; $R_{in} = \infty$; $R_{out} = r_{o1} \parallel R_{out2}$
G. Common Gate	7. $A_V = -g_{m1}\left[r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2}\right)\right]$; $R_{in} = \infty$; $R_{out} = r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2}\right)$
H. CS with Bypass Capacitor	8. $R'_i = R_G$; $A_V = -g_m R'_i$; $R'_o = r_{ds} \parallel R_D$
I. CS without Bypass Capacitor	9. $R_i = \infty$; $R'_i = R_G$; $A_V = -\frac{g_m R'_i}{1+g_m R_S}$; $R_o = r_{ds} + (1+\mu)R_S$; $\mu = g_m r_{ds}$

(c) Show that removing the emitter bypass capacitor reduces the voltage gain of a CE amplifier. Draw a voltage-divider-biased RC-coupled CE amplifier with (R_s = 400 Ω), bias resistors of 70 KΩ and 30 KΩ, (R_C = 4 KΩ), and (R_E = 1 KΩ) (bypassed). The output has an 8 KΩ load in parallel with a 750-pF capacitor. Given g_m = 50 m℧, r_π = 2 kΩ, C_π = 10 pF, C_μ = 1 pF and $r_o = \infty$. determine (i) the effective input and output capacitances using Miller's theorem and (ii) the upper cutoff frequency (f_H).

P.T.O.

Q2. **(a)** For the CE amplifier shown in the Fig. 1, the BJT has $\beta = 100$, $V_{BE} = 0.7V$, and $V_A = \infty$. Calculate (i) the DC operating point, (ii) hybrid- pi parameters, and (iii) small-signal voltage gain. If the same CE amplifier is intentionally converted into a CB amplifier, calculate the T-model parameters, small-signal voltage gain, and input resistance.

(b) In the circuit shown in Fig. 2, the transistors are identical. Identify type of current mirror (i) Find an expression for I_{out} in terms of I_{in}. (ii) If $\beta = 100$ and $V_{BE} = 0.7$ V, calculate the output current. (iii) If the resistance R_C is varied from 1 kΩ to 4 kΩ, calculate the range of the output current.

(c) Calculate the small-signal voltage gain of the amplifier shown in Fig. 3, if the MOSFETs have: $\mu_n C_{ox} = 80$ $\mu A/V^2$, $\mu_p C_{ox} = 40$ $\mu A/V^2$,$(W/L)_N = 15$, $(W/L)_P = 30$, $\lambda_n = 0.01$ V^{-1}, $\lambda_p = 0.02$ V^{-1} and $I_{Bias} = 0.2$ mA

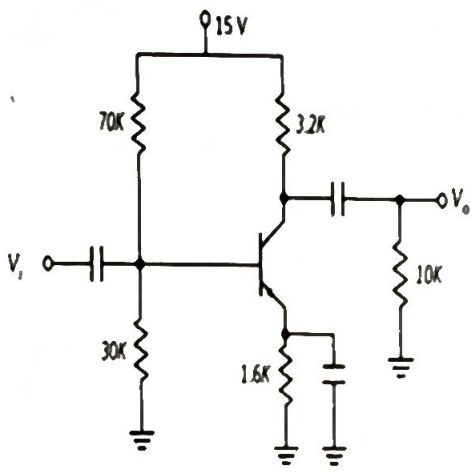


Fig. 1 CE amplifier

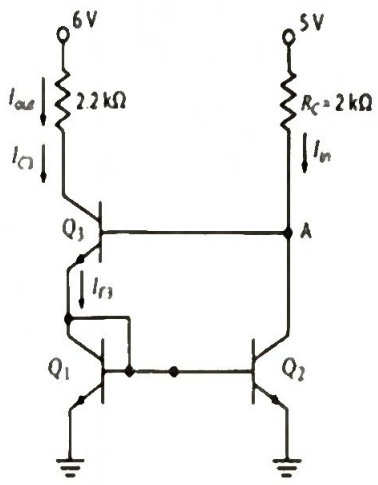


Fig. 2 Current mirror

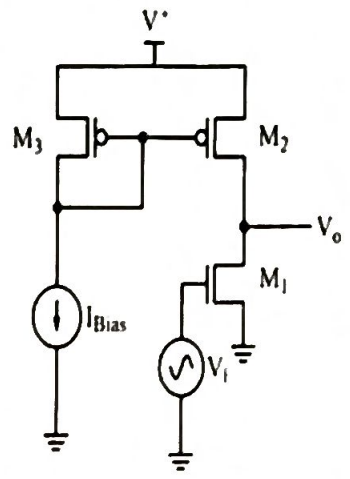


Fig. 3 CMOS common-source amplifier (M_1) with PMOS current-mirror load (M_2, M_3)

3+4+3

Q3. (i) Draw the internal schematic of OPAMP IC 741. Design logarithmic amplifier using MOSFET in feedback whether it works as log amplifier or square root amplifier. Explain working of instrumentation amplifier.

(ii) Consider the three operational amplifier circuits shown in the figure. Assuming ideal operational amplifiers and ideal diodes, answer the following: **(a)** For Fig. 4, the input voltages are $v_{i1} = (1+2\sin \omega t)$ mV and $v_{i2} = -10$ mV. Determine the output voltage v_O **(b)** For Fig. 5, the input voltage is vi=1.5 V. Determine the current i_o in the indicated direction. **(c)** For Fig. 6, determine the output voltage v_o for v_i=+2 V and v_i= -2 V respectively.

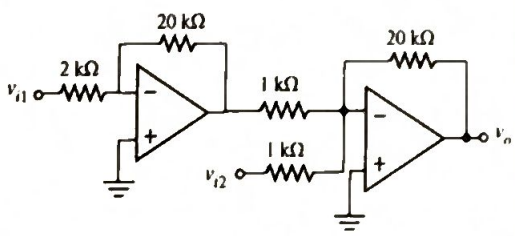


Fig. 4

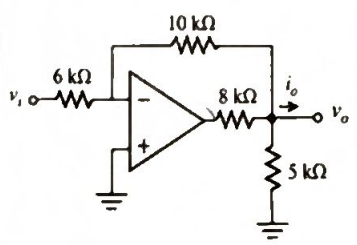


Fig. 5

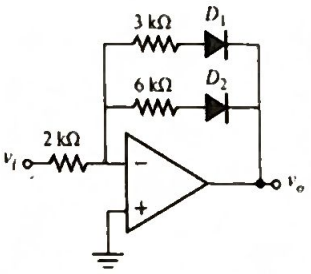


Fig. 6

6+4

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